Module 5

Q. Planning, Monitoring, and Controlling Cycle

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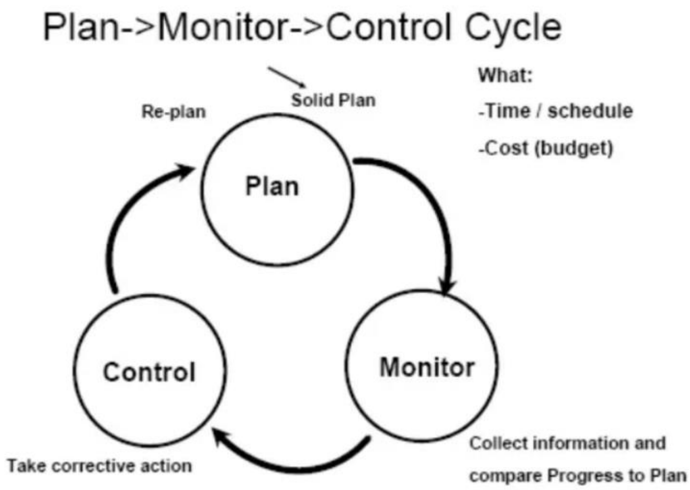

1. The planning, monitoring, and controlling cycle is an important part of project management.
2. It helps ensure that the project is completed according to scope, time, cost, and quality requirements.
3. This cycle is continuous throughout the project lifecycle.
4. It helps project managers compare actual project performance with planned performance.
5. If any difference is found, corrective actions are taken to keep the project on track.
6. The cycle mainly consists of three stages: Planning, Monitoring, and Controlling.

1. Planning

1. Planning is the first stage of the cycle.
2. In this stage, the project goals and objectives are clearly defined.
3. The project scope, schedule, and budget are prepared.
4. Tasks are identified and assigned to team members.
5. Resources such as manpower, equipment, and materials are planned.
6. Risks are identified and strategies are prepared to manage them.
7. Planning provides a roadmap for completing the project successfully.
8. Good planning helps reduce confusion and delays.

Example

9. In a software development project, planning includes deciding the features, development schedule, testing process, and required developers.

2. Monitoring

1. Monitoring is the process of tracking project progress regularly.

2. It helps check whether project activities are being completed according to the plan.
3. Project managers compare actual performance with planned performance.
4. Monitoring helps identify delays, cost overruns, and quality issues.
5. It also helps identify risks and problems at an early stage.
6. Project reports, meetings, and progress reviews are used for monitoring.
7. Continuous monitoring helps maintain project efficiency.

Example

8. In a construction project, the project manager monitors whether the building work is progressing according to the schedule.

3. Controlling

1. Controlling is the process of taking corrective actions when deviations occur.
2. If the project is behind schedule, extra resources may be added.
3. If project costs increase, unnecessary expenses may be reduced.
4. Controlling helps ensure that the project remains within scope, time, and budget.
5. It also helps maintain the required quality standards.
6. Proper control prevents project failure and improves project success.

Example

7. In a website development project, if testing takes longer than planned, the project manager may assign additional testers to complete the work faster.

Steps in the Planning, Monitoring, and Controlling Cycle

1. Define project goals and objectives.
2. Prepare project schedule, budget, and resource plan.
3. Assign tasks to team members.
4. Monitor project progress regularly.
5. Compare actual performance with planned performance.
6. Identify deviations and problems.
7. Take corrective actions to solve the problems.
8. Continue monitoring until the project is completed.

Importance of Planning, Monitoring, and Controlling Cycle

1. It helps complete projects on time.
2. It helps control project costs.
3. It improves resource utilization.
4. It helps manage risks effectively.
5. It improves communication among project team members.
6. It increases the chances of project success.

Q. Earned Value Management (EVM) Techniques for Measuring Value of Work Completed

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1. Earned Value Management (EVM) is a project management technique used to measure project performance.
2. EVM helps compare the actual work completed with the planned work and actual cost.
3. It helps project managers understand whether the project is ahead or behind schedule.
4. It also helps determine whether the project is under budget or over budget.
5. EVM combines scope, time, and cost in a single system.
6. It is widely used for monitoring project progress and performance.

Main Components of Earned Value Management

1. Planned Value (PV)

1. Planned Value is the estimated value of work that should have been completed by a certain date.
2. It is also called Budgeted Cost of Work Scheduled (BCWS).
3. Planned Value represents the planned budget for the scheduled work.

Example

4. Suppose a project has a total budget of ₹1,00,000.
5. According to the schedule, 50% of the work should be completed in one month.
6. Therefore, the Planned Value after one month will be ₹50,000.

2. Earned Value (EV)

1. Earned Value is the value of the actual work completed up to a certain date.
2. It is also called Budgeted Cost of Work Performed (BCWP).
3. Earned Value shows how much work has actually been completed in terms of budget.

Example

4. Suppose only 40% of the work is completed after one month.
5. If the total budget is ₹1,00,000, the Earned Value will be ₹40,000.

3. Actual Cost (AC)

1. Actual Cost is the actual amount of money spent on the completed work.
2. It is also called Actual Cost of Work Performed (ACWP).
3. Actual Cost shows how much money has actually been used.

Example

4. Suppose the project has spent ₹45,000 to complete 40% of the work.
5. Therefore, the Actual Cost is ₹45,000.

Important EVM Formulas

1. Schedule Variance (SV)

1. Schedule Variance shows whether the project is ahead or behind schedule.
2. Formula:
3. SV = EV − PV
4. If SV is positive, the project is ahead of schedule.
5. If SV is negative, the project is behind schedule.

Example

6. EV = ₹40,000 and PV = ₹50,000
7. SV = ₹40,000 − ₹50,000 = −₹10,000
8. This means the project is behind schedule.

2. Cost Variance (CV)

1. Cost Variance shows whether the project is under budget or over budget.
2. Formula:
3. CV = EV − AC
4. If CV is positive, the project is under budget.
5. If CV is negative, the project is over budget.

Example

6. EV = ₹40,000 and AC = ₹45,000
7. CV = ₹40,000 − ₹45,000 = −₹5,000
8. This means the project is over budget.

3. Schedule Performance Index (SPI)

1. SPI measures schedule efficiency.
2. Formula:
3. SPI = EV / PV
4. If SPI is greater than 1, the project is ahead of schedule.
5. If SPI is less than 1, the project is behind schedule.

Example

6. SPI = ₹40,000 / ₹50,000 = 0.8
7. This means the project is progressing slower than planned.

4. Cost Performance Index (CPI)

1. CPI measures cost efficiency.

2. Formula:
3. CPI = EV / AC
4. If CPI is greater than 1, the project is under budget.
5. If CPI is less than 1, the project is over budget.

Example

6. CPI = ₹40,000 / ₹45,000 = 0.89
7. This means the project is spending more money than planned.

Advantages of Earned Value Management

1. EVM helps measure project performance accurately.
2. It combines cost, schedule, and scope in one system.
3. It helps identify project problems early.
4. It improves decision making and project control.
5. It helps forecast future project performance.

Q. Team Management

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1. Team management is the process of leading, organizing, and coordinating team members to achieve project goals.
2. It involves planning team activities, assigning responsibilities, and monitoring performance.
3. Good team management helps improve productivity and project success.
4. Effective team management ensures that team members work together efficiently.
5. It helps reduce conflicts and misunderstandings in the project team.
6. Team management is an important responsibility of the project manager.

Objectives of Team Management

1. The main objective of team management is to achieve project goals successfully.
2. It helps improve coordination among team members.
3. It ensures proper use of team skills and abilities.
4. It helps maintain discipline and teamwork in the project.
5. It improves communication and trust among team members.

Functions of Team Management

1. Planning Team Activities

1. Team management involves planning project tasks and activities.
2. The project manager decides which tasks should be assigned to each team member.
3. Proper planning helps avoid confusion and delays.

2. Assigning Roles and Responsibilities

1. Every team member should have a clear role and responsibility.
2. Proper role assignment helps ensure that all tasks are completed efficiently.
3. Team members should understand their duties clearly.

3. Communication Management

1. Communication is an important part of team management.
2. Project managers should encourage regular communication among team members.
3. Meetings, emails, and reports help improve communication.
4. Good communication reduces misunderstandings and errors.

4. Motivation of Team Members

1. Team management involves motivating team members to perform better.
2. Motivation improves employee confidence and job satisfaction.
3. Rewards, appreciation, and recognition help motivate team members.

4. A motivated team works more efficiently.

5. Conflict Management

1. Conflicts may occur in a project due to different opinions or misunderstandings.
2. Team management helps identify and resolve conflicts quickly.
3. Proper conflict management improves teamwork and project performance.

6. Performance Monitoring

1. Project managers regularly monitor the performance of team members.
2. Performance monitoring helps identify strengths and weaknesses.
3. It also helps improve overall team productivity.

Characteristics of an Effective Team

1. Team members should communicate openly with each other.
2. Team members should cooperate and support each other.
3. There should be trust and respect among team members.
4. Team members should focus on common project goals.
5. Team members should be willing to solve problems together.

Example of Team Management

1. In a software development project, the project manager assigns developers for coding, testers for testing, and designers for interface design.
2. The project manager conducts regular meetings to track project progress.
3. The project manager also resolves conflicts and motivates team members to complete the project on time.

Q. Scope Creep

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1. Scope creep refers to the uncontrolled increase in the project scope during project execution.
2. It occurs when new tasks, features, or requirements are added without proper approval.
3. Scope creep usually happens after the project has already started.
4. These changes are often made without increasing the project budget, time, or resources.
5. Scope creep can lead to project delays, increased costs, and reduced quality.
6. Therefore, scope creep is considered a major problem in project management.

Causes of Scope Creep

1. Unclear project requirements can cause scope creep.
2. Poor communication between stakeholders and the project team may lead to misunderstandings.
3. Clients may request additional features during the project.
4. Lack of proper documentation can also create confusion about the project scope.
5. Weak project management and lack of change control may increase scope creep.
6. Stakeholders may change their expectations during project execution.

Effects of Scope Creep

1. Scope creep increases the total amount of work in the project.
2. It may delay the project schedule.
3. It can increase project costs because extra work requires more resources.
4. Scope creep may create pressure on the project team.
5. It can reduce the quality of the final product if the team rushes to complete additional work.
6. It may also create conflicts between team members and stakeholders.

Example of Scope Creep

1. A company initially requests a simple website with login and payment features.
2. During development, the client asks for additional features such as live chat, customer reviews, and AI recommendations.
3. These extra features increase the project workload and delay the project timeline.
4. This situation is an example of scope creep.

Methods to Control Scope Creep

1. The project scope should be clearly defined at the beginning of the project.
2. All project requirements should be documented properly.
3. Stakeholders should agree on the project scope before project execution starts.
4. A formal change control system should be used to manage changes.
5. Regular communication should be maintained between the project team and stakeholders.
6. Project managers should monitor the project scope continuously.

Q. Project Audit

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1. A project audit is a systematic review of a project to evaluate its performance and progress.
2. It helps determine whether the project is meeting its objectives, budget, schedule, and quality standards.
3. A project audit is conducted during or after the completion of the project.
4. It helps identify strengths, weaknesses, risks, and problems in the project.
5. Project audits help organizations improve future project performance.
6. It is an important tool for project monitoring and control.

Objectives of Project Audit

1. The main objective of a project audit is to evaluate project performance.
2. It helps determine whether the project is progressing according to plan.
3. It helps identify project risks and problems.
4. It ensures that project resources are being used efficiently.
5. It helps improve communication and coordination among team members.
6. It also helps document lessons learned for future projects.

Types of Project Audit

1. Initial Audit

1. An initial audit is conducted at the beginning of the project.
2. It checks whether the project plan, scope, budget, and schedule are realistic.
3. It helps identify possible risks before project execution begins.

2. In-Process Audit

1. An in-process audit is conducted while the project is running.
2. It checks project progress, cost, schedule, and quality.
3. It helps identify problems during the project so corrective actions can be taken.

3. Final Audit

1. A final audit is conducted after the project is completed.
2. It evaluates whether the project achieved its objectives.
3. It checks the final cost, schedule, quality, and customer satisfaction.
4. It also records lessons learned from the project.

Areas Covered in Project Audit

1. Project scope is reviewed to check whether all deliverables were completed.
2. Project schedule is reviewed to see if the project was completed on time.
3. Project cost is reviewed to check whether the project stayed within budget.

4. Quality standards are reviewed to ensure that the final output meets requirements.
5. Risk management practices are also evaluated.

Steps in Project Audit

1. The project data and documents are collected.
2. Project plans, reports, and schedules are reviewed.
3. Interviews may be conducted with team members and stakeholders.
4. The project performance is compared with the original plan.
5. Problems and deviations are identified.
6. Suggestions are provided to improve future projects.

Example of Project Audit

1. A company completes a software development project.
2. After project completion, the management reviews whether the software was delivered on time and within budget.
3. The company also checks whether the customer is satisfied with the software quality.
4. This review process is called a project audit.

Advantages of Project Audit

1. Project audit helps identify project strengths and weaknesses.
2. It improves future project planning and management.
3. It helps reduce risks in future projects.
4. It ensures better use of resources.
5. It improves accountability and transparency.